APARTMENT MANAGEMENT SYSTEM (AMS)

Product Documentation & End-to-End User Demonstration

Application	AMS desktop application
Technology	C++ / Qt 6 / QML, using a model-view structure
Document purpose	Explain why each screen exists, what it does, and how to demonstrate the complete workflow.
Scope	Flat Master, Meter Reading, Bill Generation, and View Bills

This document provides an overview of the Apartment Management System (AMS), its key features, screen functionality, and end-to-end operational workflow. It serves as a user guide for understanding and demonstrating the application.

1. Product Overview

The Apartment Management System (AMS) is a desktop application intended to support a small apartment/community's flat records, monthly electricity-meter readings, and bill generation. It provides a single user interface for maintaining flat information, recording readings by month and year, generating bills, reviewing bill details, and exporting selected bills to PDF.

The main window has a navigation panel on the left and a working area on the right. Selecting a navigation item switches the working area to the corresponding screen.

1.1 Main user workflow

Step	Screen	Why it comes here
1	Flat Master	Create the flat records that identify the units in the property.
2	Meter Reading	Record each flat's meter reading for a selected month and year.
3	Bill Generation	Calculate and create bills using readings and the configured rate/maintenance values.
4	View Bills	Review, filter, select, delete, or export generated bills.

Important sequence: Flat records should exist before meter readings are entered. Meter readings should be available for the relevant billing month before generating bills. After generation, use View Bills to inspect and export the results.

1.2 Intended users

- Apartment administrator or association staff maintaining flat details.
- Staff recording monthly meter readings.
- Staff generating and distributing bill PDFs.

1.3 Current documented scope

The supplied project contains four main navigation screens: Flat Master, Meter Reading, Bill Generation, and View Bills. Bill payment tracking and monthly expense entry/maintenance were discussed as possible future features, but they are not included as existing features in this documentation because the supplied screen code does not establish that they have been implemented.

2. Main Window and Navigation

Why this screen exists

The main window acts as the application shell. It keeps the four work areas accessible from one place, avoiding the need to open separate application windows for each task.

What it contains

- Application title: “Apartment Management System”.
- Left navigation list: Flat Master, Meter Reading, Bill Generation, and View Bills.
- Right-side content area: displays the selected screen.

How navigation works

1. Launch AMS.

The main application window opens.

2. Choose a section in the left navigation panel.

Click Flat Master, Meter Reading, Bill Generation, or View Bills.

3. Work in the selected screen.

The main content area changes to the selected section.

4. Move to another section when ready.

Click another navigation item; the application switches the content area without requiring a separate window.

Navigation map

Flat Master	→	**Meter Reading**	→	**Bill Generation**	→	**View Bills**

This diagram represents the recommended business workflow. The screens can be opened in any order from the navigation panel, but the data should be prepared in the order shown for a meaningful billing run.

3. Flat Master Screen

Why this screen exists

Flat Master establishes the list of flats that AMS can refer to when capturing meter readings and preparing bills. It is the basic reference-data screen for the apartment.

What it does

- Accepts a Flat Number.
- Accepts a Flat Type (used to distinguish flat categories such as 1BHK, 2BHK, or 3BHK where applicable).
- Accepts a Status value for the flat.
- Displays existing flat records in a table with Flat No, Flat Type, and Status columns.
- Provides Add, Update, Delete, and Clear actions.
- Uses a confirmation dialog for deletion.

How to use it

1. Open Flat Master.

Select “Flat Master” from the left navigation panel.

2. Add a flat.

Enter the flat number, choose/enter the flat type, set the status, then click “Add”. Confirm the record appears in the table.

3. Select an existing flat before updating or deleting.

Click the corresponding row so the form can reflect the selected record.

4. Update a flat.

Edit the relevant fields and click “Update”. Check that the table shows the revised details.

5. Delete a flat.

Select the row and click “Delete”. Review the confirmation prompt and confirm only if the selected record should be removed.

6. Clear the form.

Click “Clear” to reset the input fields and selection for the next operation.

Expected result

The flat table should show the current records and their flat type/status. These flat records provide the reference list for the Meter Reading workflow.

User guidance

Use a consistent flat-number format throughout the application. Before deleting a flat, verify that the selected row is the intended one. If a flat should remain available for historical billing, consider whether changing its status is more appropriate than deleting it.

4. Meter Reading Screen

Why this screen exists

A bill requires a meter reading associated with a flat and a billing period. This screen records that reading against a flat number, month, and year, and provides a table for reviewing existing entries.

What it does

- Accepts Flat Number, Month, Year, and Meter Reading.
- Displays saved entries in a table with Flat No, Month, Year, and Reading columns.
- Provides Add, Update, Delete, and Clear actions.
- Uses a confirmation dialog when deleting a selected reading.
- Provides a selected-row interaction so an existing reading can be edited or removed.

How to use it

1. Open Meter Reading.

Select “Meter Reading” in the left navigation panel.

2. Choose the flat.

Select or enter the flat number for which the reading is being recorded.

3. Set the billing period.

Enter the month and year to which the meter reading belongs.

4. Enter the meter reading.

Enter the meter reading value for that flat and period.

5. Save the entry.

Click “Add” and verify the record appears in the table.

6. Correct an existing entry if needed.

Select the entry, change the relevant fields, and click “Update”. Verify the updated values.

7. Remove an incorrect entry if needed.

Select the entry, click “Delete”, and confirm the deletion.

8. Prepare for the next entry.

Use “Clear” to reset the form.

Data quality checks for the operator

- Confirm that the flat number is correct.
- Confirm that month and year match the intended billing cycle.
- Enter the meter reading as recorded from the meter; avoid confusing the meter reading with units consumed.
- Check for duplicate entries for the same flat and billing period before proceeding.

Expected result

The reading table should contain the flat’s reading for the selected period. Bill Generation uses the reading data as an input to calculate consumption and bill amounts.

5. Bill Generation Screen

Why this screen exists

This screen brings together the billing period and the rate/maintenance inputs needed to generate bills. It presents the generated bills in a table so the operator can review the results before using the View Bills screen for further actions.

Inputs visible in the screen

Input	Purpose
Bill Month	Month for which bills are generated.
Bill Year	Year for which bills are generated.
Rate Per Unit	Electricity rate used to calculate the electricity charge. The screen's displayed example/default is 8.50; verify the value before each run.
Maintenance (1BHK)	Maintenance amount applied to the relevant flat category. The screen's displayed example/default is 1500.
Maintenance (2BHK)	Maintenance amount applied to the relevant flat category. The screen's displayed example/default is 2000.
Maintenance (3BHK)	Maintenance amount applied to the relevant flat category. The screen's displayed example/default is 2500.

Generated bill table

The table displays Flat, Month, Year, Previous Reading, Current Reading, Units, Rate, Electricity, Maintenance, and Total. This lets the operator inspect the reading inputs and calculated amounts together.

How to generate bills

1. Open Bill Generation.

Select "Bill Generation" in the left navigation panel.

2. Set the period.

Enter the target Bill Month and Bill Year.

3. Verify the financial inputs.

Check Rate Per Unit and the maintenance values for 1BHK, 2BHK, and 3BHK. Do not assume the displayed defaults are correct for the current period.

4. Click "Generate Bills".

The application runs the bill-generation operation using the available flat and meter-reading data.

5. Read the operation result.

The screen reports the outcome, including the number of bills generated and any flats skipped where applicable.

6. Review the generated table.

Check flat, period, readings, units, rate, electricity amount, maintenance amount, and total amount.

7. Proceed to View Bills.

Use the left navigation panel to inspect, filter, select, and export bills.

Operational note

The exact calculation rules and validation behaviour are implemented in the bill manager. Operators should verify the generated values against a known example before using the system for live billing, especially when changing rates or maintenance amounts.

6. View Bills Screen

Why this screen exists

View Bills is the review and action area for generated bills. It lets the operator locate bills, select one or more records, export selected bills to PDF, or delete selected bills.

What it displays

The list includes Flat, Month, Year, Previous, Current, Units, Rate, Electricity, Maintenance, and Total. A row checkbox indicates selection; selected rows are visually highlighted. A header checkbox selects all rows or clears all selections, and its checked state reflects whether all displayed bills are selected.

Available actions

- Search/filter bills using Flat No, Month, and Year fields.
- Click “Search” to apply the entered filter values.
- Click “Clear Filters” to remove the active filter criteria.
- Select individual bill rows using the row checkboxes.
- Use the header checkbox to select all bills or clear all selections.
- Click “Export Selected PDFs” to export the selected bill records to PDF.
- Click “Delete Selected Bills” to delete selected bills after confirming the deletion prompt.

How to use it

1. Open View Bills.

Select “View Bills” in the left navigation panel.

2. Find a bill.

Enter any available Flat No, Month, and/or Year criteria, then click “Search”.

3. Reset the search.

Click “Clear Filters” to return to the unfiltered bill list.

4. Select one bill.

Tick the checkbox on that bill’s row. The row should be highlighted.

5. Select multiple bills.

Tick each required row, or tick the header checkbox to select all bills. Untick the header checkbox to clear all selections.

6. Export selected bills.

Click “Export Selected PDFs”. Follow any file/save behaviour provided by the application and verify the reported export result.

7. Delete selected bills.

Click “Delete Selected Bills”. Review the confirmation dialog carefully and confirm only if the selected bills should be removed.

PDF export

The current implementation builds a PDF filename from the flat number, bill month, and bill year (for example, Bill_102_10_2026.pdf). The precise output location depends on the application’s current working directory or the path logic in the bill manager. After export, verify the destination and open a sample PDF to check its contents.

Deletion caution

Deleting a bill is a destructive action. Before confirming, review the selected count and the visible rows.
If multiple bills are selected, the action applies to all selected records.

7. End-to-End Demonstration Script

This script is designed for a live demo to a user, tester, or stakeholder. Use sample/test data rather than live resident data during demonstrations.

Demo setup

- Prepare one or more test flats with different flat types.
- Have known meter readings for a selected month and year.
- Choose a rate per unit and maintenance amounts suitable for the demonstration.
- Ensure the test period does not unintentionally overwrite or duplicate live billing data.

Demonstration flow

1. Start at the main window.

Point out the application title and four navigation choices. Explain that each section is displayed in the right-hand content area.

2. Create a test flat.

Open Flat Master, add a flat such as “DEMO-101”, choose a flat type and status, and show the record in the table.

3. Record a meter reading.

Open Meter Reading, select DEMO-101, choose a test month/year, enter the reading, click Add, and show the saved row.

4. Generate the bill.

Open Bill Generation, set the same billing period, verify the rate and maintenance amounts, then click Generate Bills.

5. Explain the result.

Show the generated count and any skipped-flat message. Point to previous reading, current reading, units consumed, rate, electricity charge, maintenance charge, and total.

6. Locate the bill.

Open View Bills and search using the demo flat number and billing period.

7. Demonstrate selection.

Select the demo bill’s checkbox and show the highlighted row. If multiple test bills exist, demonstrate the header checkbox to select all and clear all.

8. Export to PDF.

Click Export Selected PDFs, then locate and open the resulting PDF. Confirm that the flat, period, and bill amounts are readable.

9. Demonstrate filters.

Search by flat/month/year and then use Clear Filters to restore the full list.

10. Demonstrate delete only with disposable test data.

Select a demo bill, click Delete Selected Bills, review the confirmation prompt, and confirm. Avoid deleting live records during a presentation.

What the demo proves

- Flat records can be maintained.
- Meter readings can be recorded against a flat and billing period.

- Bill generation uses the billing period, readings, rate, and maintenance inputs.
- Generated bills can be reviewed and filtered.
- One or more bills can be selected for PDF export or deletion.

8. Common Questions and Troubleshooting

A flat does not appear in the reading workflow

Return to Flat Master and confirm that the flat record exists and its details are correct. Then reopen Meter Reading.

A bill is skipped during generation

Review the operation result message and check whether the flat has a meter reading for the selected month/year and whether the required flat data exists. The precise skip conditions are governed by the bill-generation logic.

The header checkbox changes state unexpectedly

The header checkbox reflects the selection state of the bill list. If any row is unselected, the all-selected state should be false. Select or clear rows and confirm the header state updates consistently.

A PDF is not where expected

Check the application's current working directory and the path used by the PDF export implementation. Export one test bill and confirm the exact destination before running a batch export.

9. Current Scope and Not-yet-documented Features

The supplied code and the project discussion establish the following core features: Flat Master maintenance, Meter Reading maintenance, bill generation, bill listing/filtering, selection, deletion, and PDF export. The following features were discussed as possible additions but should be treated as future scope unless separately implemented and tested:

- Recording whether a bill has been paid or remains unpaid.
- Monthly expense entry, editing, and deletion.
- Reports or dashboards for payment status and monthly expenses.

Do not present these future ideas as existing capabilities in a user demo until the implementation is completed and verified.

10. Glossary

Term	Meaning in AMS
Flat Master	Reference list of apartment flats and their type/status.
Meter Reading	Meter value recorded for a flat and a particular month/year.
Units	Electricity consumption derived from the relevant readings by the billing logic.
Rate Per Unit	Rate used to calculate the electricity charge.
Maintenance	Maintenance charge selected according to the flat category and configured amount.
Generated Bill	Bill record created for a flat and billing period.
PDF Export	Export of selected bill records into PDF files.

Product Documentation: This document describes the functionality and operational workflows of the Apartment Management System. For deployment and operational use, ensure that the application is configured correctly and that the expected functionality is verified in the target environment.